bridge, and collisions of a child's marbles. In Chapter 2 we briefly met an example of a Newtonian law, the inverse square law for gravity.

By 1915 Einstein and others had found strong evidence that the Newtonian laws fail in the realm of the very fast (objects that move at nearly the speed of light), the realm of the very large (our universe as a whole), and the realm of intense gravity (for example, black holes). To remedy these failures Einstein gave us his revolutionary *relativistic laws of physics* (Figure 3.2). Using the concepts of warped time and warped space (which I describe in the next chapter), the relativistic laws predicted and explained the expansion of the universe, black holes, neutron stars, and wormholes.